

Pharmacology of Antidepressants and Lithium



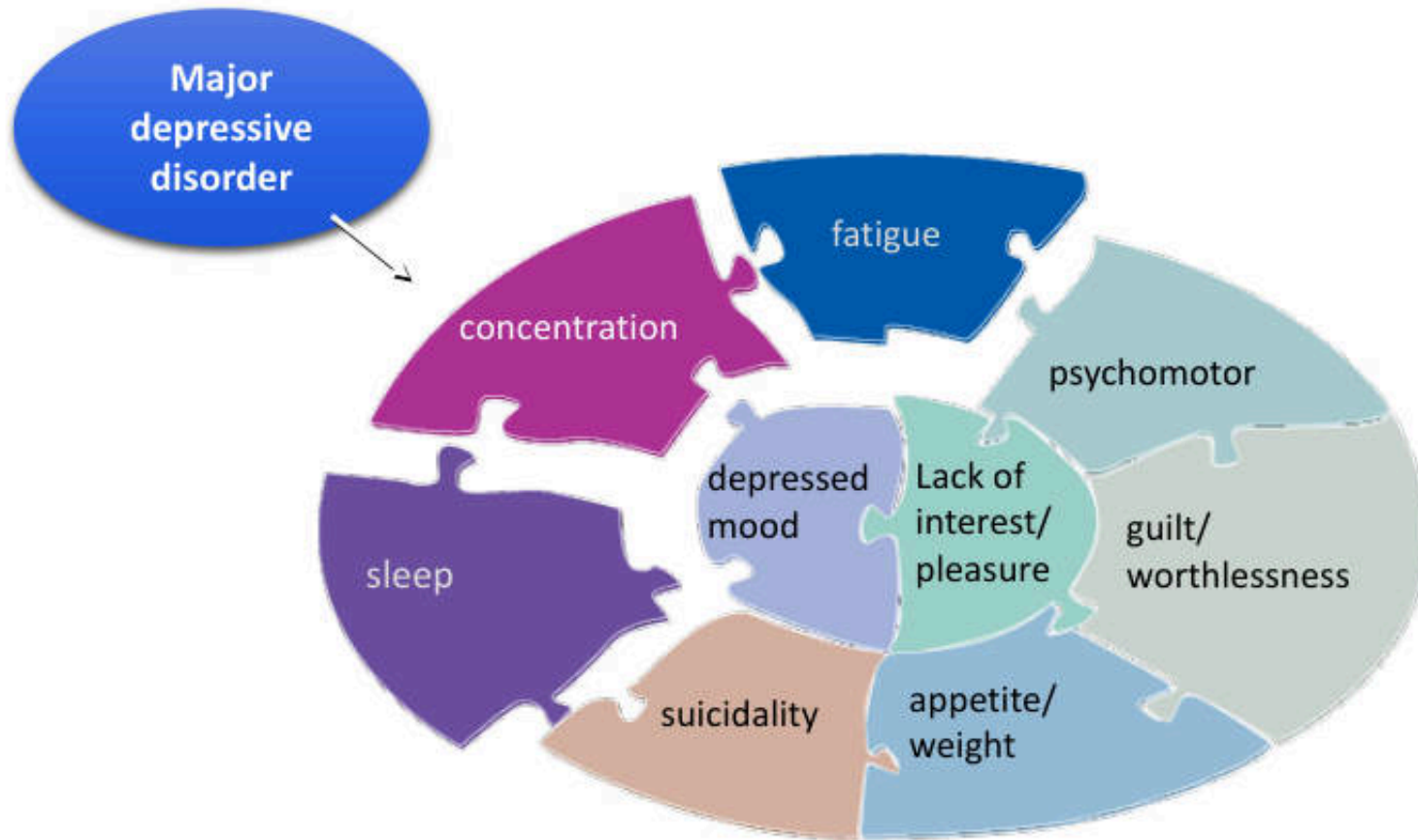
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วัตถุประสงค์เชิงพฤติกรรม

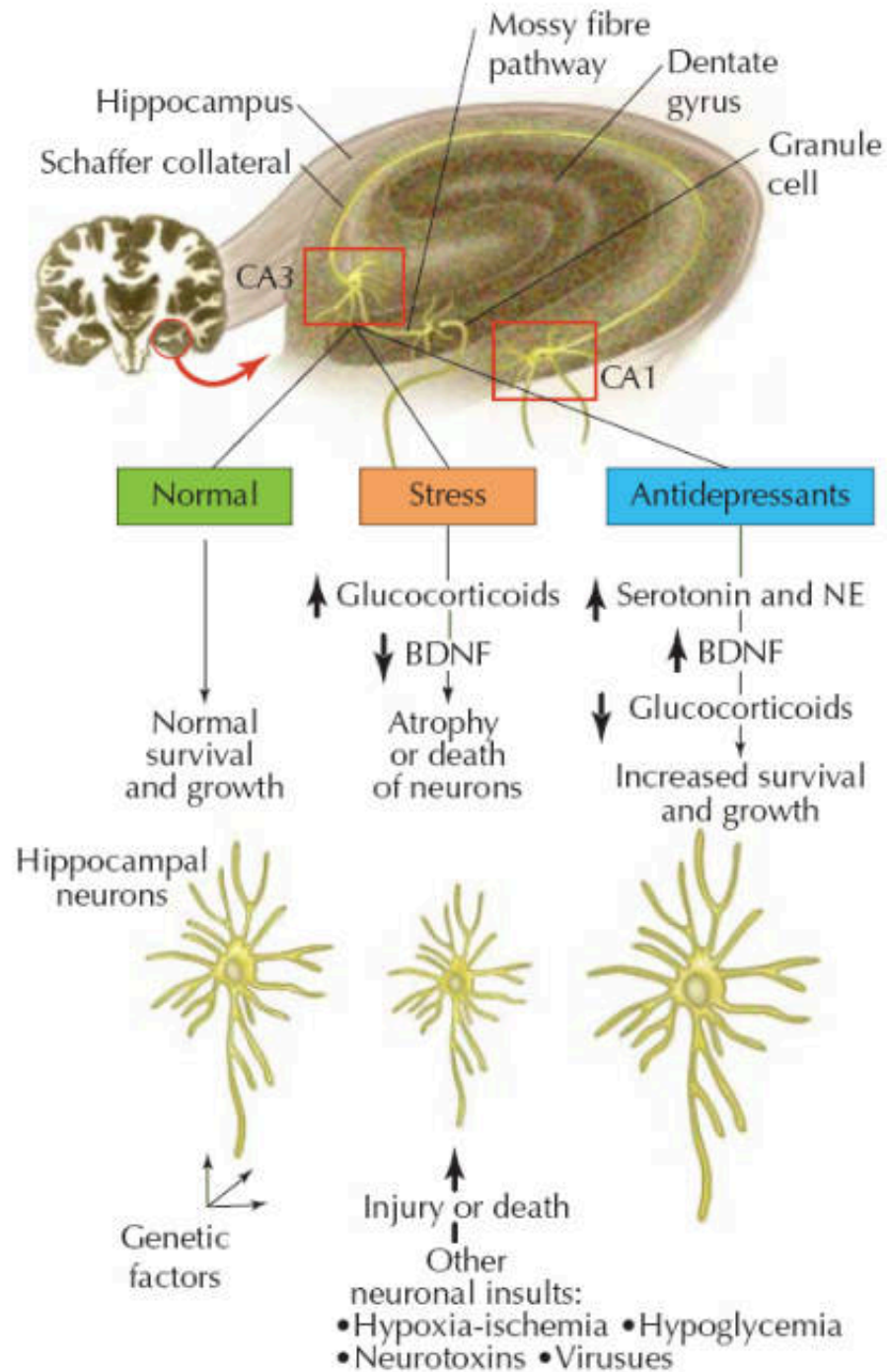
- * อธิบายผลของ antidepressants ต่อการเกิด neurogenesis และ synaptogenesis ของ hippocampus ได้
- * จัดแบ่งกลุ่มยา antidepressants ได้ตามกลไกการออกฤทธิ์และอธิบายผลทางเภสัชวิทยาของ antidepressants ที่สำคัญได้
- * เปรียบเทียบความแตกต่าง antidepressants กลุ่มต่างๆ ที่สำคัญได้ ทั้งในแง่ผลการรักษาและอาการไม่พึงประสงค์
- * ระบุลักษณะทางเภสัชจลนศาสตร์ของ antidepressants ที่สำคัญและใช้บ่อยทางคลินิกได้
- * ระบุลักษณะทางเภสัชวิทยาที่สำคัญของ lithium ได้
- * อธิบาย drug interaction ที่สำคัญของ antidepressants และ lithium ได้

Symptoms of major depressive disorder



Pathophysiologic hypotheses of depression

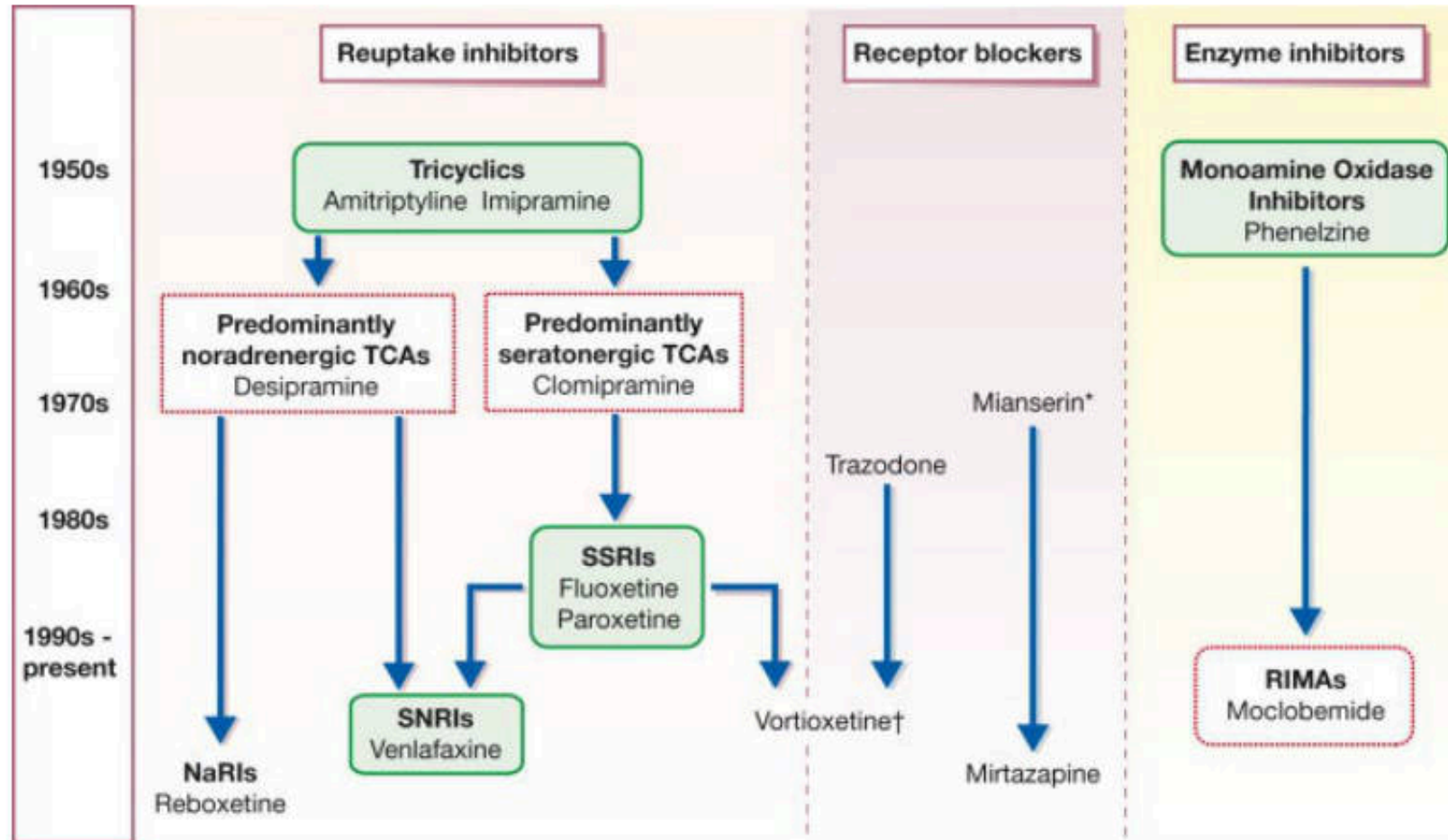
- ✿ Monoaminergic hypothesis
- ✿ Signalling adaptation hypothesis
- ✿ Neurotrophic hypothesis
 - ❖ Chronic stress-induced depression



A molecular and cellular model for the action of antidepressant treatments and the pathophysiology of stress-related disorders

BDNF = brain-derived neurotrophic factor

Development of antidepressants



Key-Drugs classes in boxed, shaded fields represent the three major antidepressants groups, tricyclics (TCAs), selective serotonin reuptake inhibitors (SSRIs) and monoamine oxidase inhibitors.

Novel compounds are left unboxed.

NaRI-noradrenaline reuptake inhibitor

SNRI-serotonin, and noradrenaline reuptake inhibitor

RIMA-reversible inhibitor of monoamine oxidase

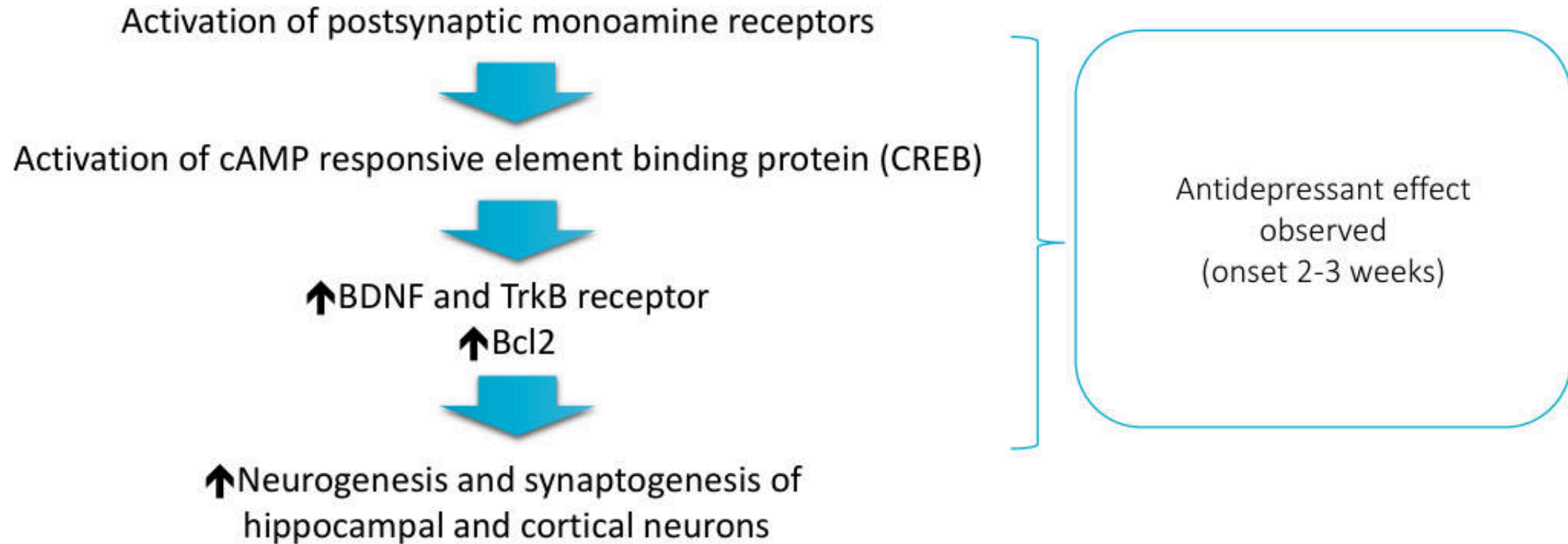
*-Mianserin is rarely used due to associations with aplastic anaemia

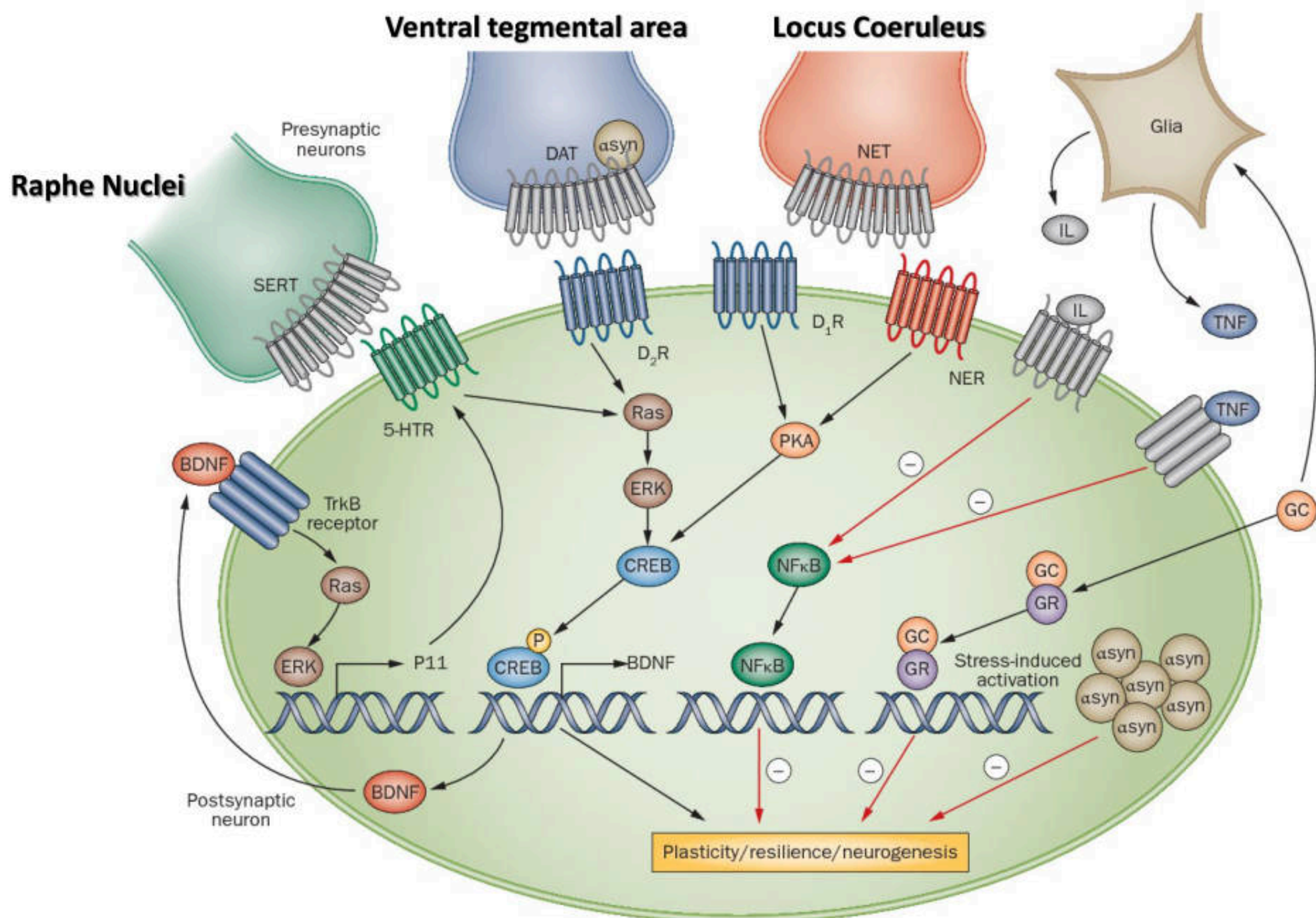
†-Vortioxetine is both a reuptake inhibitor and receptor blocker, as well as acting as an agonist at other 5HT receptors

Classification of Antidepressants

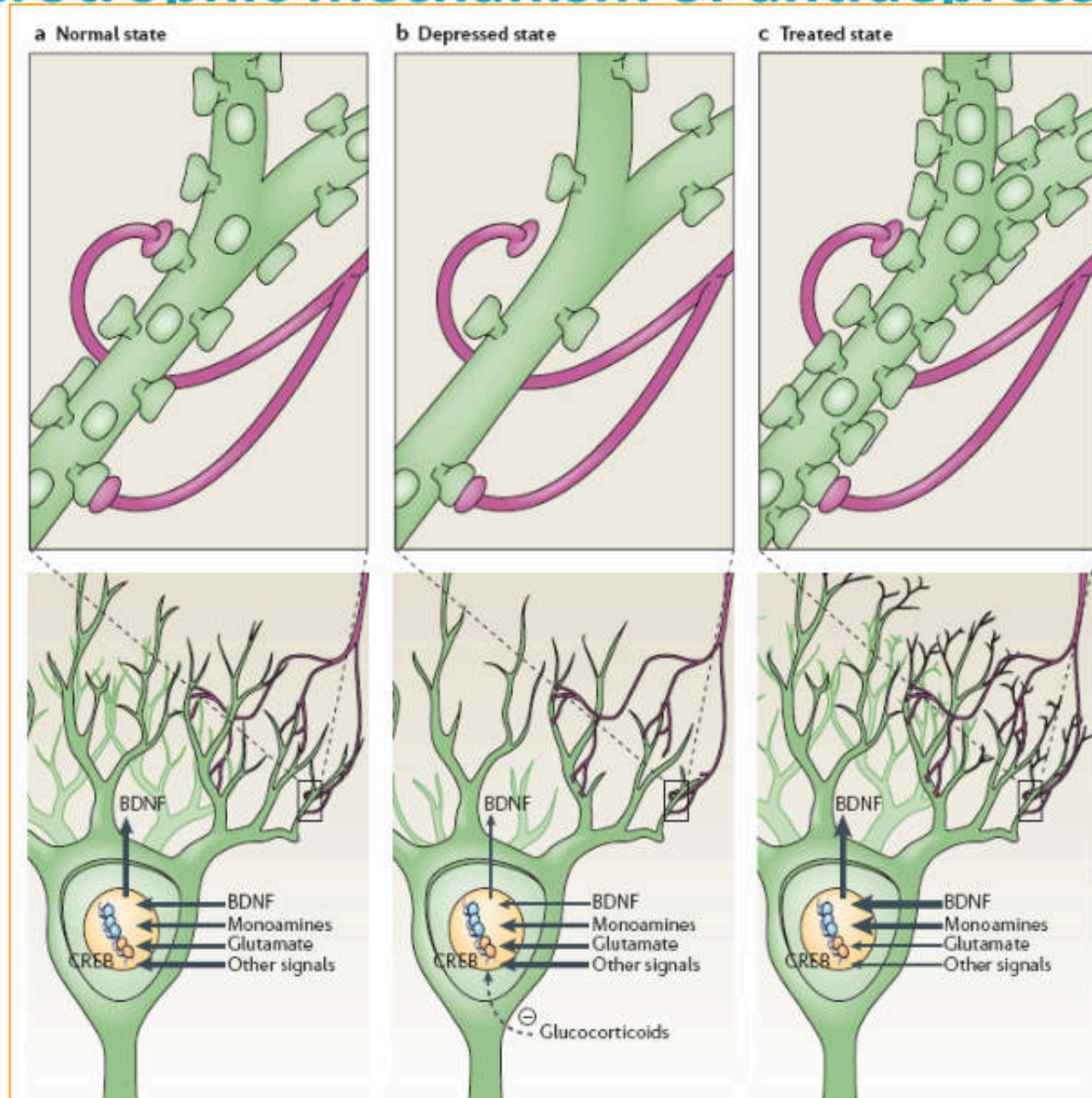
1. Monoamine oxidase inhibitors (MAOIs), reversible inhibitors of MAO-A (RIMAs)
2. Tricyclic antidepressants (TCAs)
3. Selective serotonin reuptake inhibitors (SSRIs)
4. Serotonin antagonist and reuptake inhibitor (SARIs)
5. Serotonin norepinephrine reuptake inhibitors (SNRIs)
6. Norepinephrine dopamine reuptake inhibitors (NDRIs)
7. Noradrenergic and specific serotonergic antidepressants (NaSSAs)
8. Miscellaneous
 - ❖ Melatonergic agonist and serotonergic antagonist
 - ❖ Norepinephrine reuptake inhibitors (NRIs)
 - ❖ Serotonin antagonist and reuptake enhancers (SSREs)
 - ❖ Serotonin partial agonist reuptake inhibitors (SPARIs)
 - ❖ Multimodal antidepressants
 - ❖ NMDA receptor antagonist

Long-term elevation of monoaminergic neurotransmitters induces neuroplasticity in hippocampus and prefrontal cortex





Neurotrophic mechanism of antidepressants



1. MAO inhibitors

1.1 Irreversible MAO inhibitors; tranylcypromine, iproniazid

- ❖ Block MAO irreversibly with no selectivity
- ❖ Cheese reaction - tyramine

1.2 Reversible inhibitors of MAO-A (RIMAs); moclobemide (Aurolix®)

- ❖ Block MAO-A selectively (at low dose)

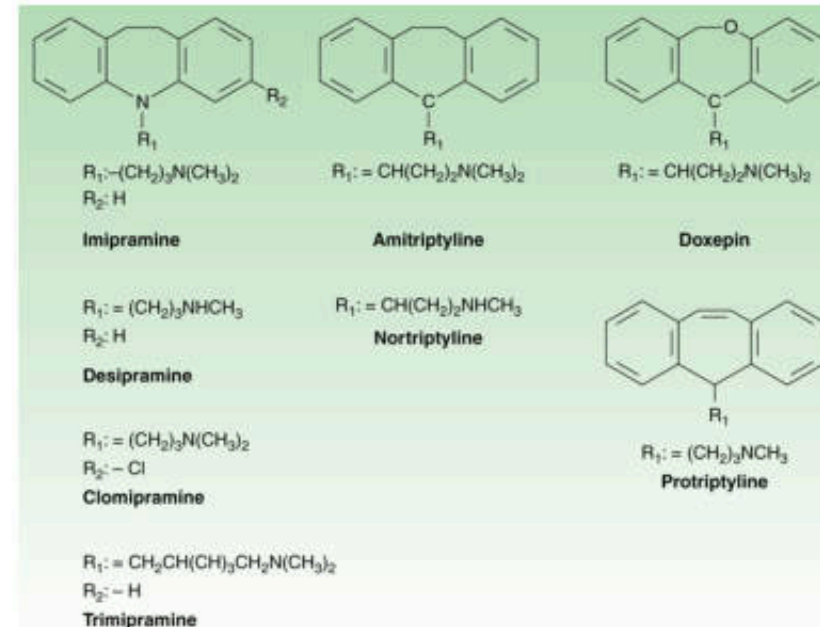
2. Tricyclic antidepressants (TCAs)

TCAs

1. Amitriptyline (Tryptanol[®])
2. Nortriptyline (Nortrilen[®])
3. Clomipramine (Trofranil[®])
4. Imipramine (Anafranil[®])
5. Maprotiline (Ludiomil[®])
6. Doxepin (Sinequan[®])
7. Melitracen in combination with flupentixol (Deanxit[®])

✧ Mechanism of action

- ✧ Block H_1 , muscarinic, $5-HT_{2A}$, α_1
- ✧ Block 5-HTT, NET
- ✧ Block Na^+ channel
- ✧ Block NMDA receptors



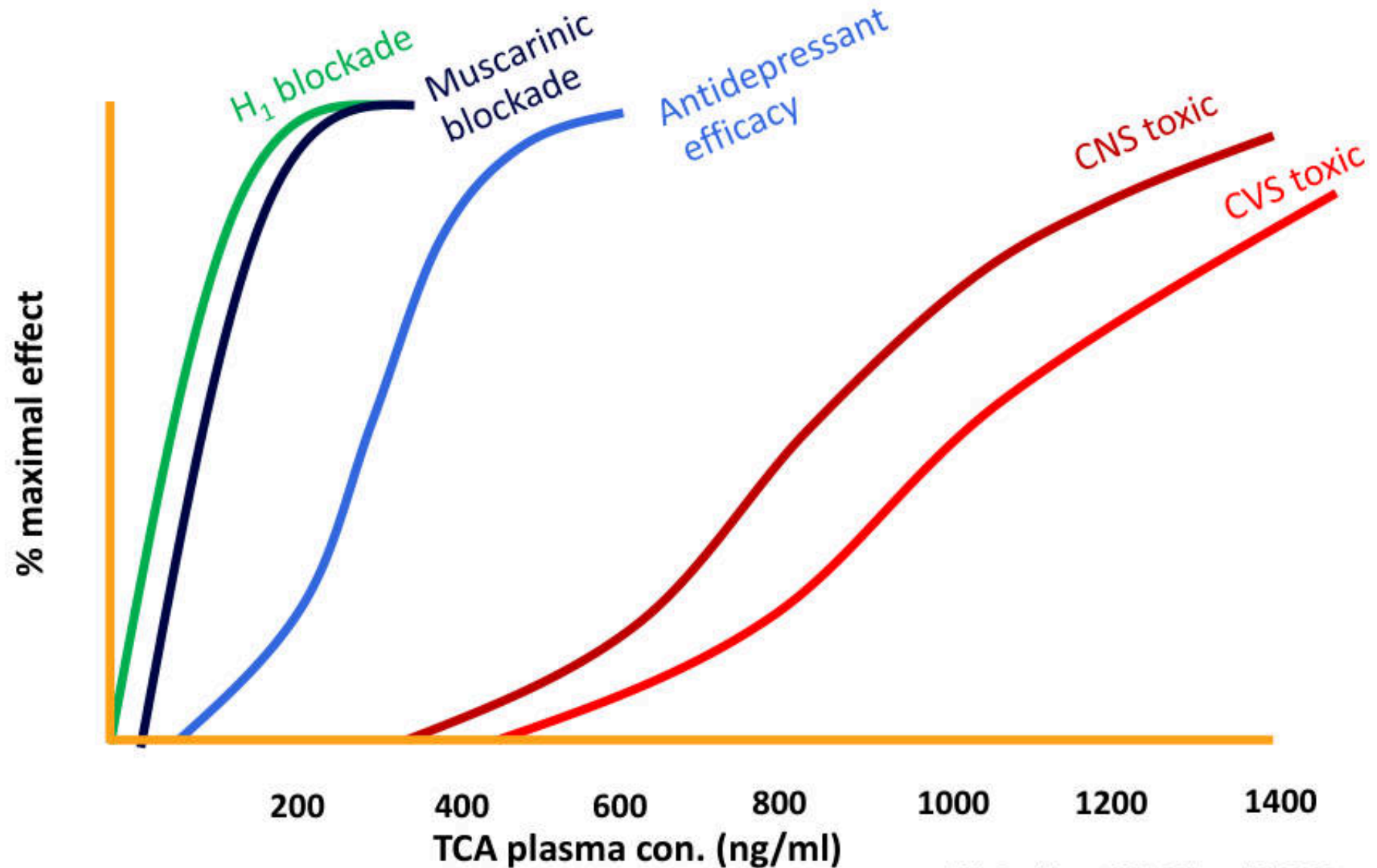
Source: Katzung BG, Masters SB, Trevor AJ: Basic & Clinical Pharmacology, 12th edition; www.accessmedicine.com

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TCA_s

Blockade of receptor/transporters	Pharmacological effects
NET	Antidepressant effect Palpitation
5-HT	Antidepressant effect Nausea/vomiting
Muscarinic (M ₁ , M ₂ , M ₃)	Anticholinergic side effects Sedation
5-HT _{2A}	Sedation, weight gain
H ₁	Sedation, weight gain
α ₁	Orthostatic hypotension
Voltage-gated sodium channels	Proarrhythmic effect, PR interval prolongation
Cardiac potassium channel (hERG)	Proarrhythmic effect, QTc interval prolongation

TCAs plasma concentration and pharmacological effects



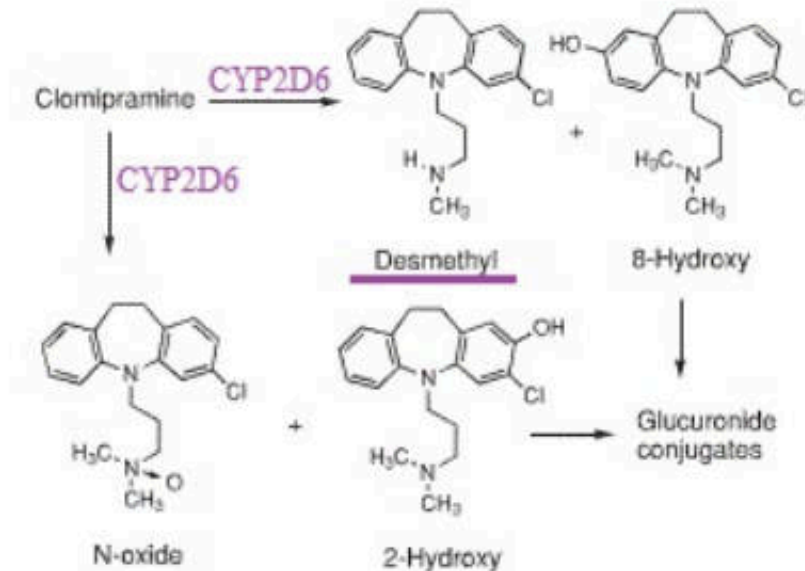
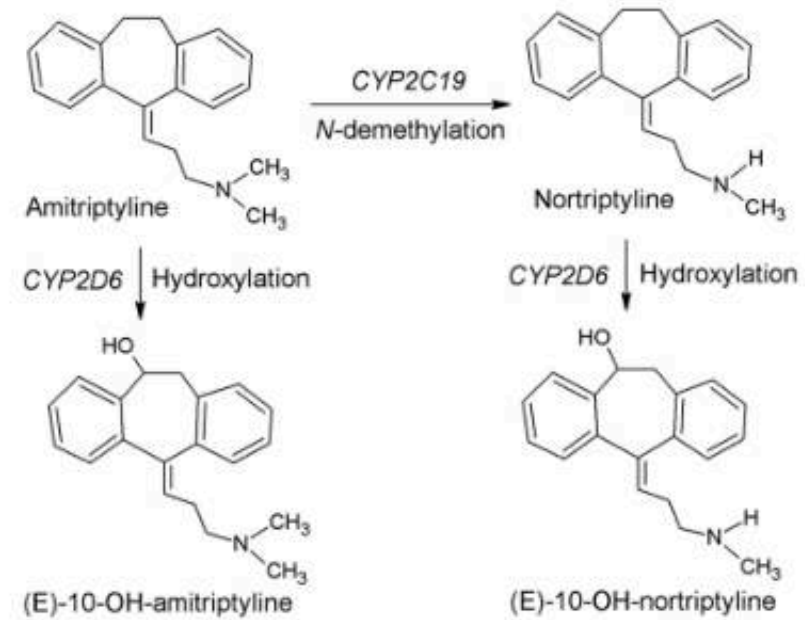
Side effects profile of commonly used TCAs

Agent	Sedation	Anticholinergic	Hypotension
Amitriptyline	++++	++++	++++
Clomipramine	+++	+++	+++
Doxepin	+++	++	++
Imipramine	+++	+++	+++
Nortriptyline	++	++	+

0/+, minimal affinity; +, mild affinity; ++, moderate affinity; +++, high affinity.

TCAs metabolism

- ✧ Most of TCAs are metabolized by CYP2D6
- ✧ Some are also metabolized by other CYPs
 - ❖ CYP2C19
 - ❖ CYP1A2



TCA safety concerns

- ✱ Overdose of TCA causes sudden death
- ✱ TCAs are associated with reduced seizure thresholds
- ✱ Elderly patients are more sensitive to TCA adverse effects
 - ❖ Reduced cardiovascular reflex
 - ❖ Sedation
 - ❖ Cognitive impairment
 - ❖ Constipation
 - ❖ Benign prostatic hypertrophy

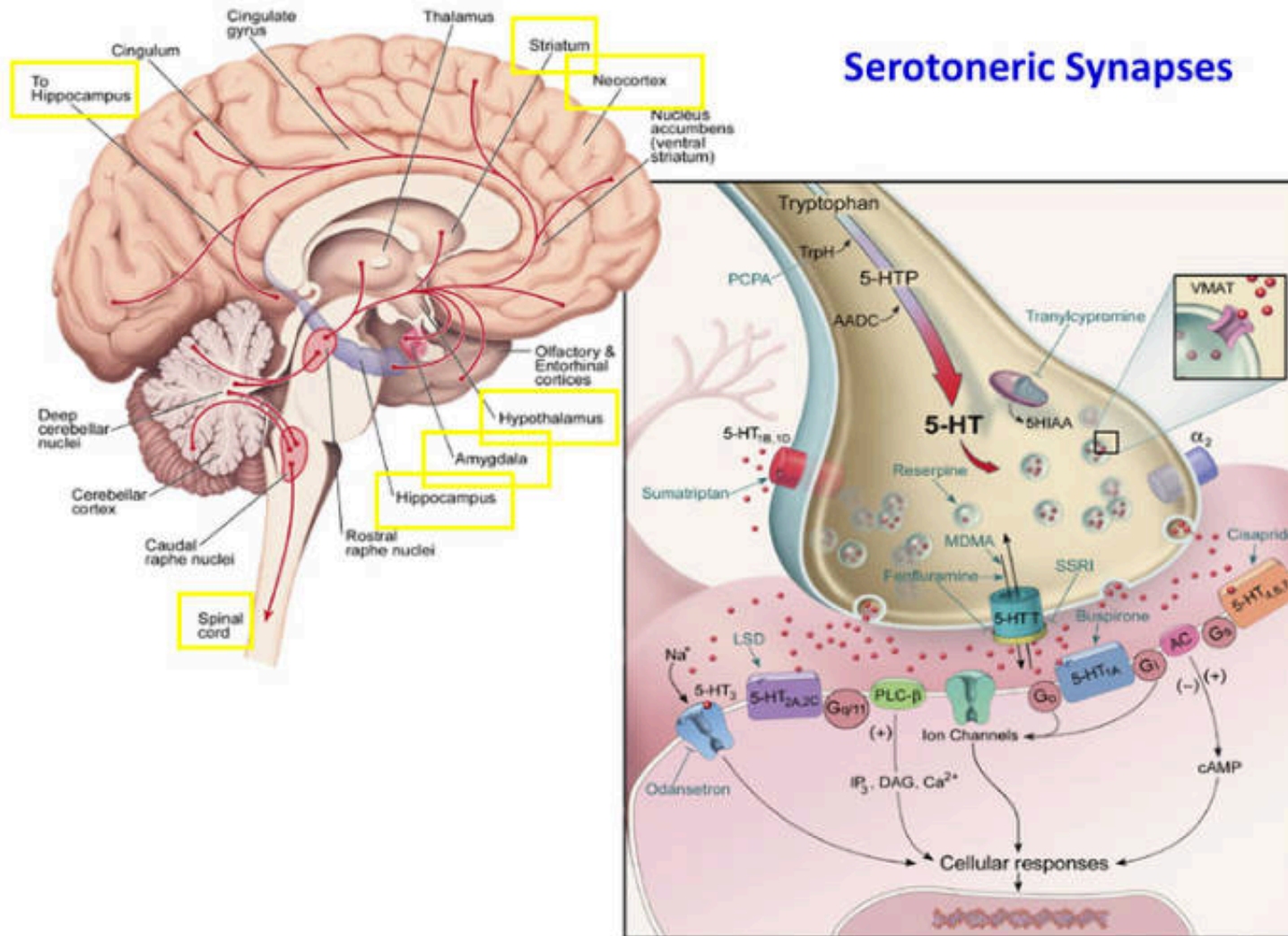
3. Selective serotonin reuptake inhibitors (SSRIs)

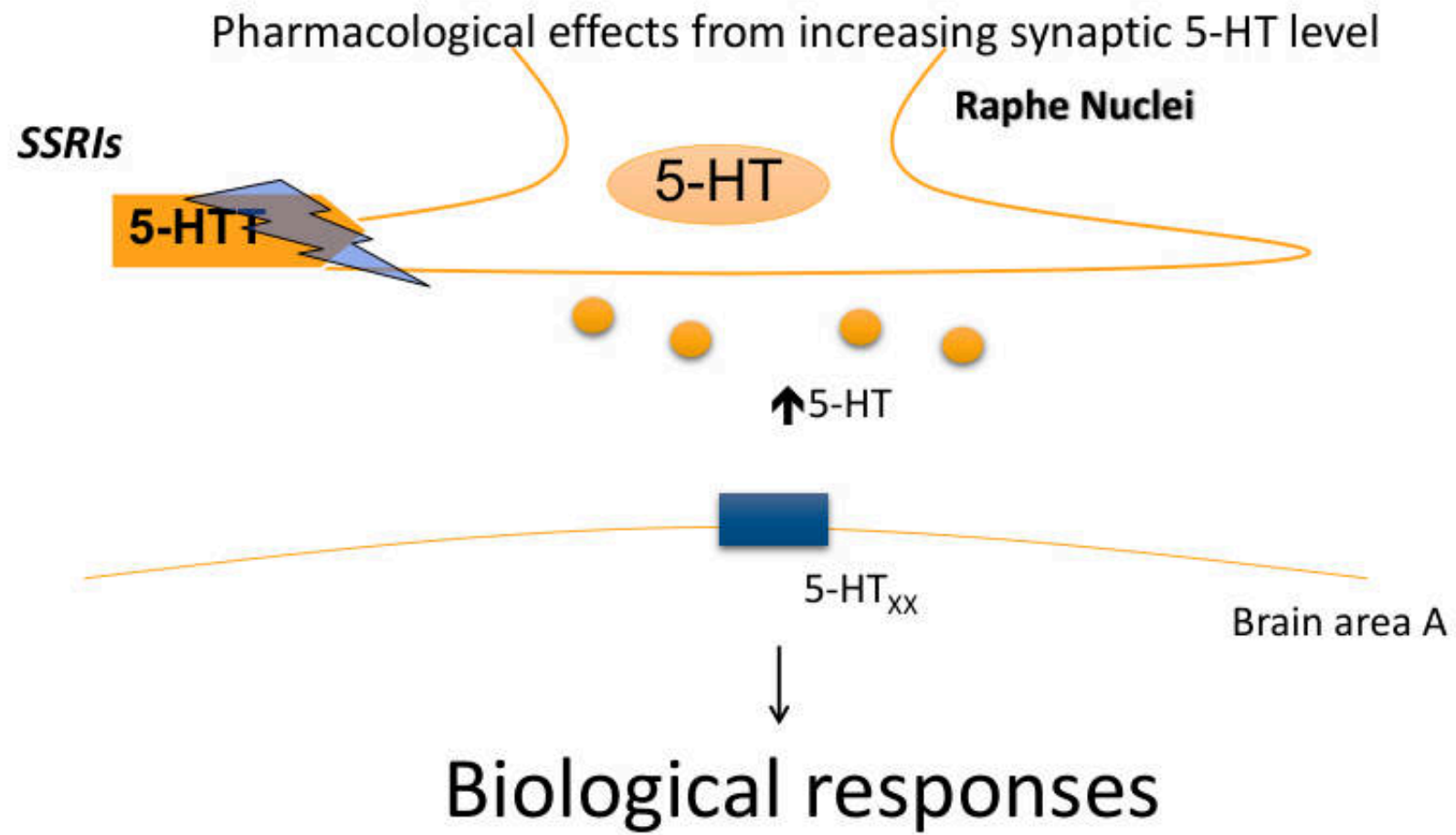
SSRIs

1. Fluoxetine (Prozac[®], Flulox[®])
2. Fluvoxamine (Faverin[®])
3. Paroxetine (Seroxat[®])
4. Sertraline (Zoloft[®])
5. Escitalopram (Lexapro[®])

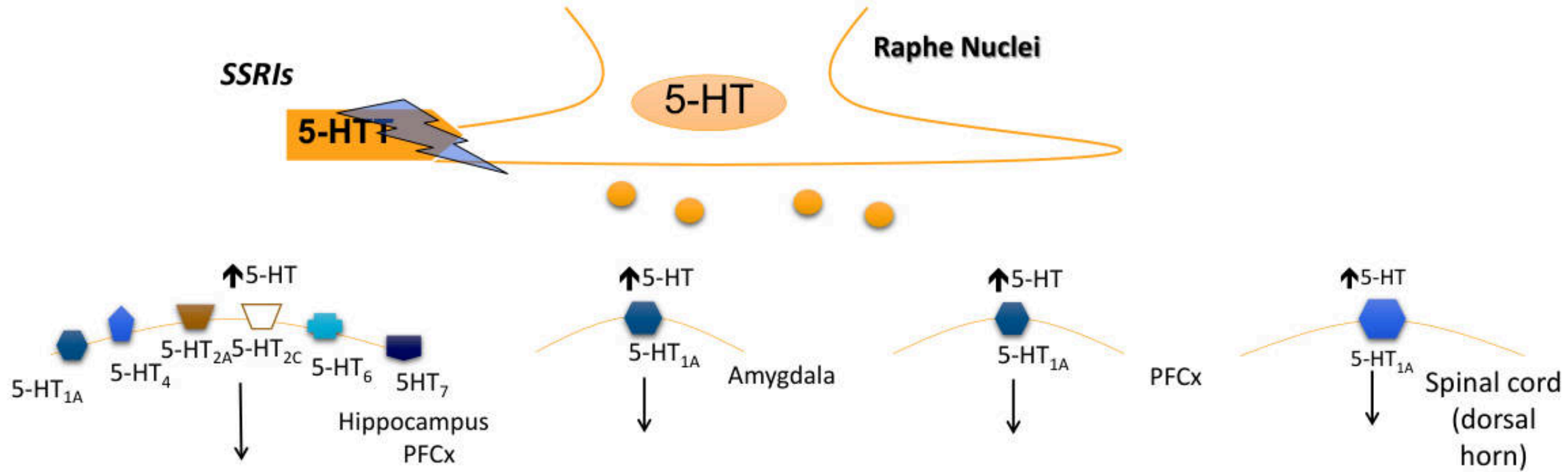
- ✧ Mechanism of action
 - ❖ Block 5-HTT with higher affinity and selectivity when compared to TCAs
- ✧ No anticholinergic effects except high dose paroxetine
- ✧ Some SSRIs precipitate CYP450 mediated drug interactions

Pharmacological effects from SSRIs





Pharmacological effects from increasing synaptic 5-HT level



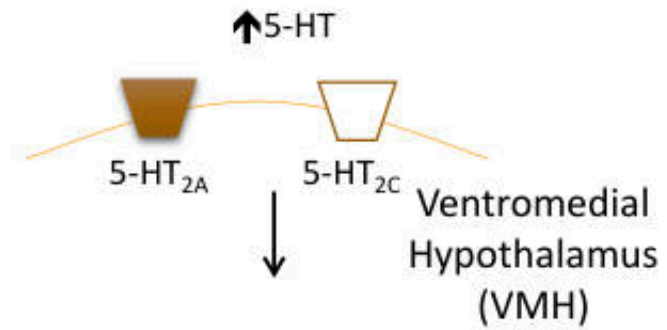
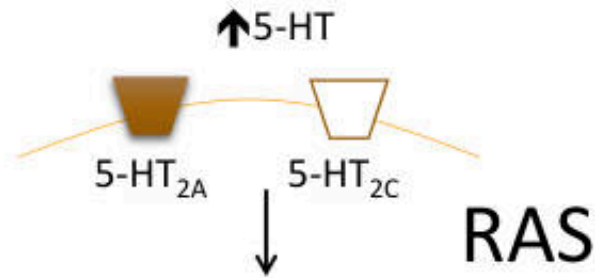
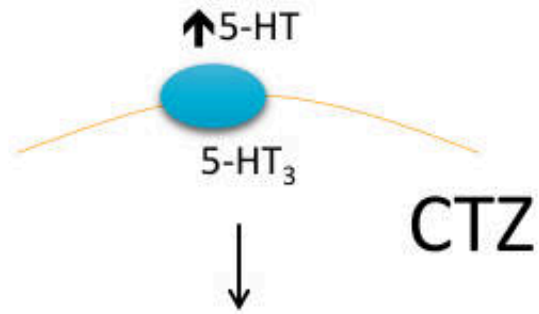
Pharmacological effects from increasing synaptic 5-HT level

Raphe Nuclei

SSRIs

5-HTT

5-HT



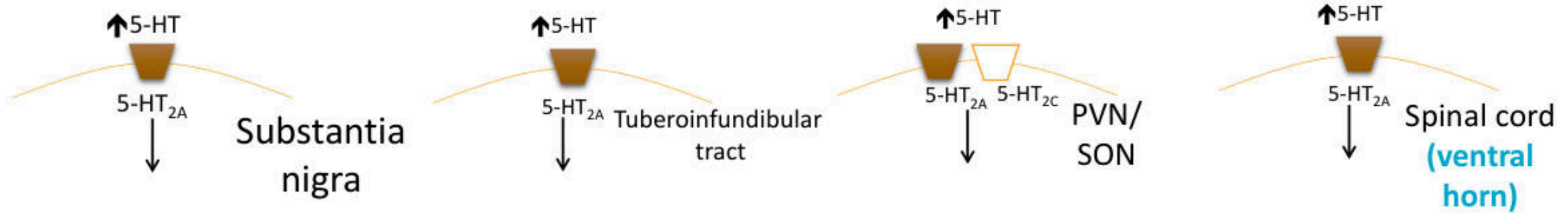
Pharmacological effects from increasing synaptic 5-HT level

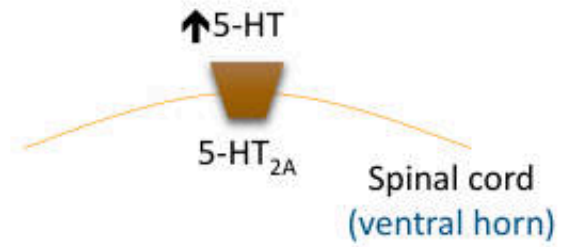
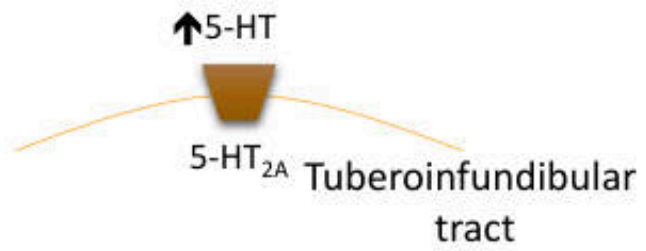
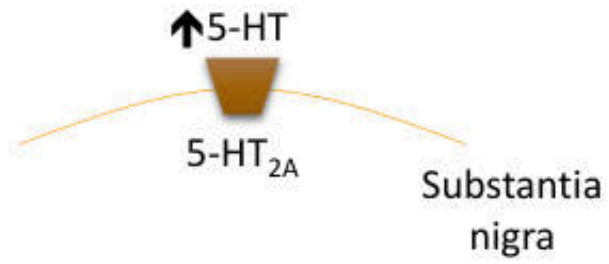
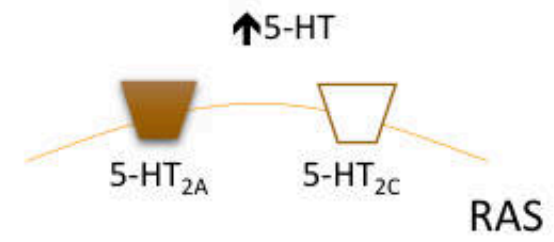
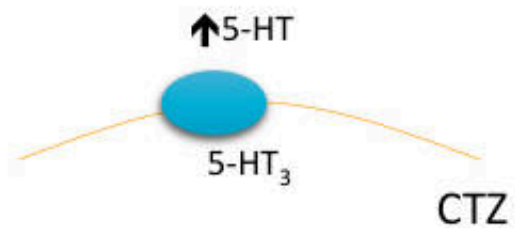
SSRIs

5-HTT

5-HT

Raphe Nuclei





Summary of side effects from blockade of 5-HTT

Receptor activation	Area	Effects
5-HT ₃	CTZ	Nausea, vomiting,
	GI tract	Nausea, vomiting, loose stool
	Blood vessel	Headache
5-HT _{2A}	Nigrostriatal tract	Akathisia and mild EPS
	Ventral horn of spinal cord	inhibit sexual reflex; orgasm, ejaculation
	Ventromedial hypothalamus (VMH)	promote satiety (+5-HT _{2C})
	Paraventricular nucleus (PVN), supraoptic nucleus (SON)	Vasopressin release (+5-HT _{2C})
	RAS	Insomnia and decrease REM sleep (+5-HT _{2C})
	Tuberoinfundibular tract	Increase prolactin release
5-HT ₄	Heart	Palpitation
5-HT _{1A}	Amygdala	Anxiolysis, anti-OCD
	Prefrontal cortex	Antidepressant
	Dorsal horn of spinal cord	Inhibit pain transmission

Pharmacokinetics of SSRIs

	Fluoxetine	Paroxetine	Sertraline	Fluoxamine	Esitalopram
Typical dose (mg/day)	10-60	20-50	50-200	50-300	20-60
Major elimination pathway	CYP2D6	CYP2D6			
Active metabolites	Norfluoxetine	-	N-desmethyl-sertraline	-	-
Elimination half-life					
Parent drug	2-3 d	21 h	26 h	15 h	35 h
Active metabolites	7-9 d	-	2-4 d	-	-
Time to steady state	30-60 d	4-5 d	4-5 d	3 d	7-8 d
Linearity	Nonlinear	Nonlinear	Linear	Nonlinear	Linear
Protein binding (%)	> 95	>95	>95	≈77	≈80
Level increased in elderly	Yes	Yes	Yes	No	No
Hepatic disease	X3	X1.8	X2	X1.6	X2
Renal disease	No	No	X1.5	No	No

Summary of the effect of antidepressant on the major human drug metabolizing cytochrome P450 enzymes

	Drug	1A2	2C9	2C19	2D6	3A3/4
SSRI	Fluoxetine	●	+++	++	+++	+
	Fluvoxamine	+++	++	+++	●	++
	Paroxetine	●	●	●	+++	●
	Sertraline	●	●	●	+	●
	Escitalopram	●	●	●	+	●
SNRI	Desvenlafaxine	●	●	●	●	●
	Venlafaxine	●	●	●	+	+
	Duloxetine	●	●	●	++	●
Other	Bupropion	?	?	?	+++	?
	Mirtazapine	●	●	●	●	●
	Vortioxetine	●	●	●	●	●

*Percent increase in plasma levels of a coadministered drug dependent on this CYP enzyme for its clearance

- No or minimal effect (<20%)
- Moderate effect (50%-150%)
- Mild effect (20%-50%)
- Substantial effect (>150%)

Modified from J Psychiatr Pract. 2020 Mar;26(2):126-134.

Serotonin syndrome

✿ Syndrome associated with excessive serotonin level

✿ Causes

- ❖ Drug-drug interactions
- ❖ Overdose of SSRIs or SNRIs
- ❖ Genetics

	CNS excitation	Mental state	Autonomic excitation	Typical cause
Serotonin toxicity (increase in CNS 5HT efflux*)				
Severe (10-100x)	Rigidity, respiratory failure	Coma	Severe hyperthermia	MAOI plus SSRI combination
Moderate (5-10x)	Opisclonus, sustained clonus, myoclonus, tremor	Confusion	Mydriasis, flushing, diaphoresis, low fever (<38.5°C)	SSRI overdose
Mild (3-5x)	Inducible clonus, hyper-reflexia	Agitation	Hypertension, tachycardia	Ecstasy use
(<3x)	Brisk reflexes	Anxiety	Nausea, diarrhoea	SSRI in therapeutic use
		Insomnia		

CNS = central nervous system; 5HT = 5-hydroxytryptamine; MAOI = monoamine oxidase inhibitor; SSRI = selective serotonin reuptake inhibitor
*Approximate extent of increase in CNS 5HT efflux seen with animal models

Serotonin withdrawal (discontinuation) symptoms

- ✿ Serotonin withdrawal (discontinuation) symptoms may occur when abruptly stopping after chronic use
- ✿ Symptoms (not life-threatening)
 - ❖ Flu-like symptoms
 - ❖ Insomnia
 - ❖ Nausea
 - ❖ Imbalance
 - ❖ Sensory disturbance
 - ❖ Hyperarousal
- ✿ Lower risk for fluoxetine

4. Serotonin antagonist and reuptake inhibitors (SARIs)

SARIs

1. Trazodone (Desirel®)

✧ Mechanism of action

- ❖ Block 5-HT_{2A/2C}, H₁
- ❖ Block 5-HTT
- ❖ Block α_1

✧ Common side effects

- ❖ Sedation, weight gain
- ❖ Orthostatic hypotension and priapism

✧ EPS, delayed ejaculation

✧ Low dose trazodone may be used as sleep aid with no anticholinergic effects

5. Serotonin norepinephrine reuptake inhibitors (SNRIs)

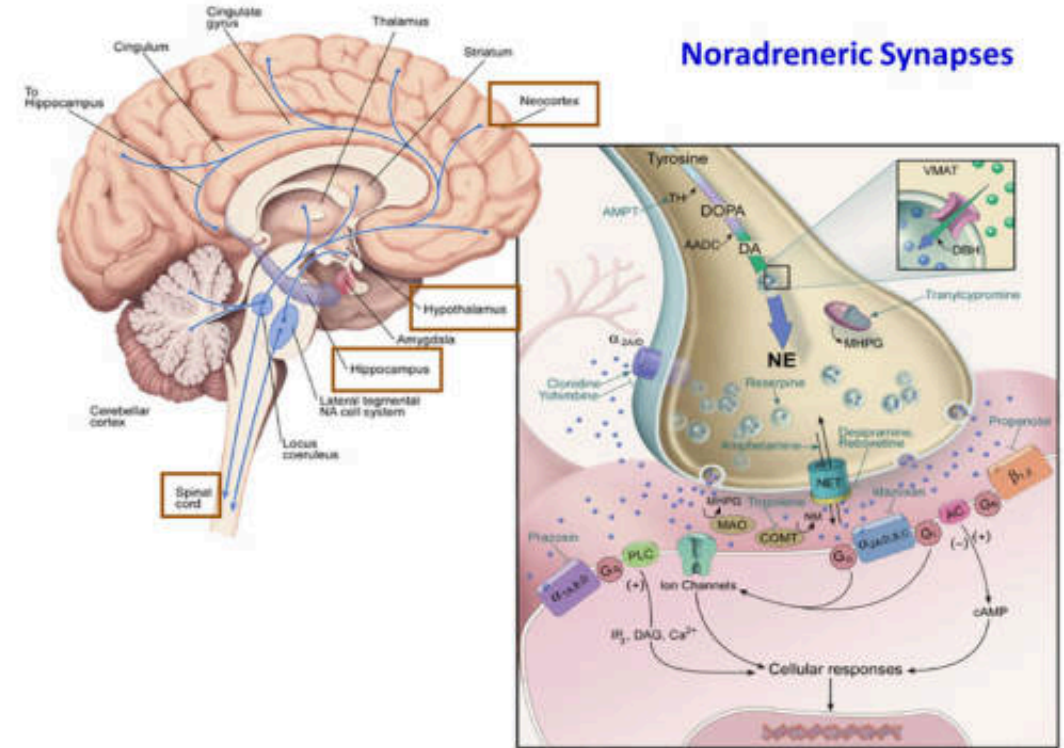
SNRIs

1. Venlafaxine (Efexor-XR[®])
2. Duloxetine (Cymbalta[®])
3. Desvenlafaxine (Pristiq[®])
4. Milnacipran (Ixel[®])
5. Levomilnacipran (Fetzima[®])

- ✧ Mechanism of action
 - ✦ Block 5-HTT and NET
- ✧ Duloxetine is a moderate inhibitor of CYP2D6

Pharmacological effects of SNRIs

- ✿ Similar to SSRIs plus noradrenergic effects
- ✿ CNS
 - ❖ Zest, energy
 - ❖ RAS
 - * Insomnia and decrease REM sleep
 - ❖ Dorsal horn of spinal cord
 - ** Inhibit pain transmission
 - ❖ Paraventricular nuclei (PVN)
 - * Appetite suppression
- ✿ Periphery: ↑ Sympathetic tone
 - ❖ ↑ BP
 - ❖ Constipation



Analgesic effect	SSRI
Insomnia	SSRI
Sympathetic side effects	SSRI
Nausea, vomiting, sexual side effect	SSRI

6. Norepinephrine dopamine reuptake inhibitors (NDRIs)

- ✧ Mechanism of action
 - ❖ Block NET and DAT (weak)
- ✧ Produce only 40-50% increase of DA level
- ✧ Beneficial in patients with cognitive or attention complaints
- ✧ Common side effects
 - ❖ Sympathetic side effects, insomnia
 - ❖ Reduced appetite
 - ❖ Sexual side effect < SSRIs, SNRIs
- ✧ Reduce seizure threshold
- ✧ Moderate-to-potent inhibitor of CYP2D6

NDRIs

Bupropion (Wellbutrin[®],
Quomem[®])

7. Noradrenergic and specific serotonergic antidepressant (NaSSAs)



Mechanism of action

- ❖ Block H_1 receptor
- ❖ Block α_2 receptor, 5-HT_{2A}, 5-HT_{2C}
- ❖ Mirtazapine block 5-HT₃
- ❖ Mianserin block α_1

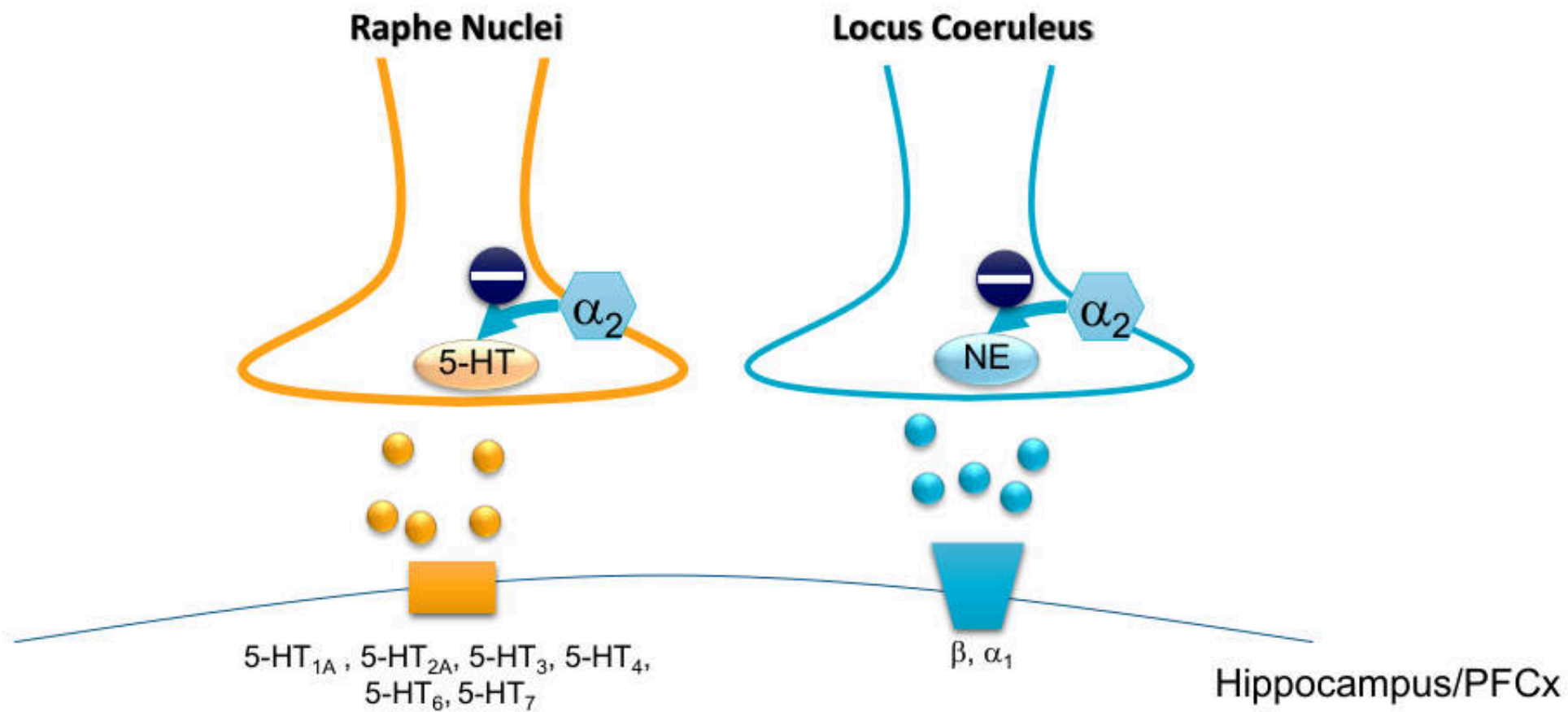


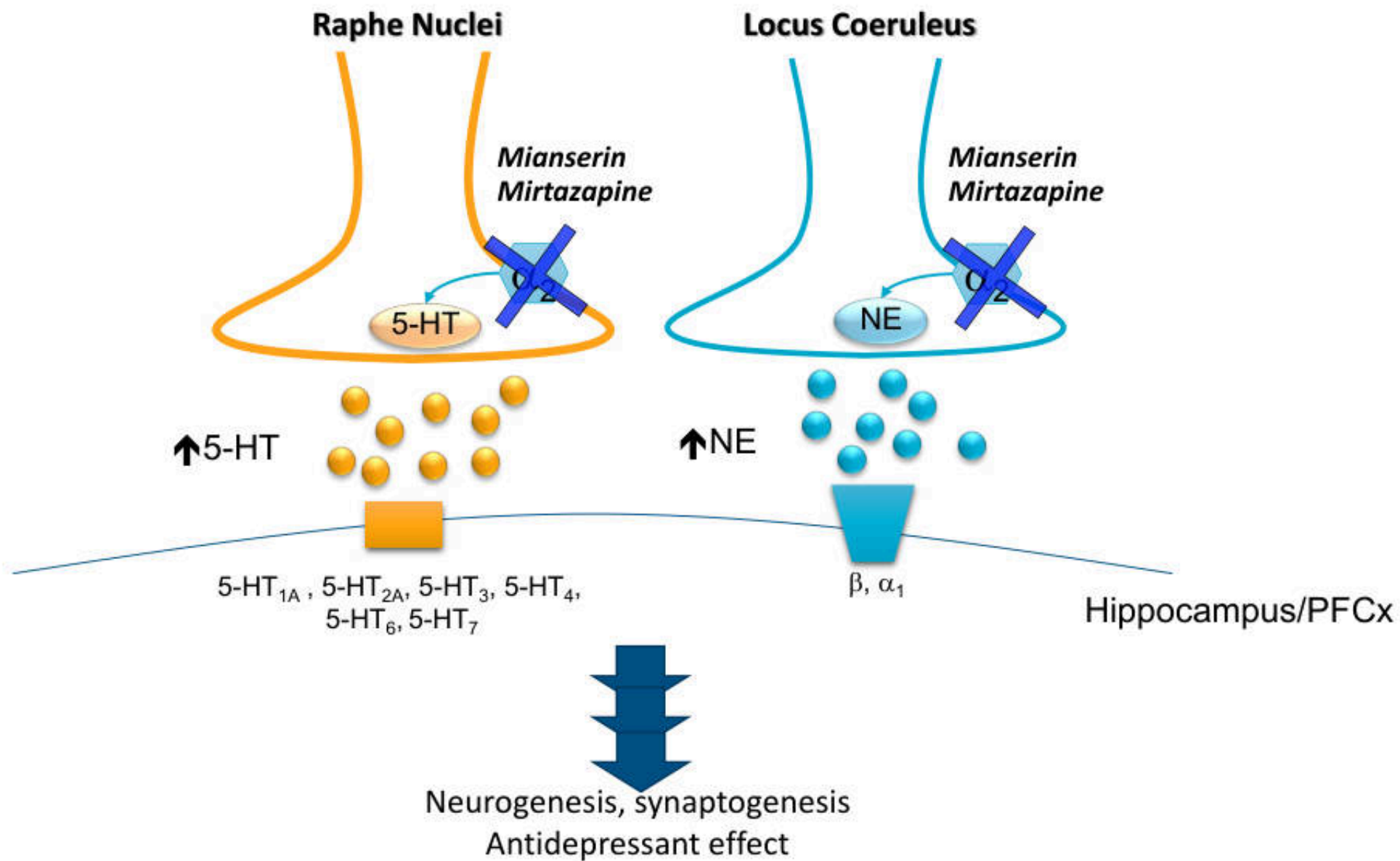
Common side effects

- ❖ Sedation, weight gain
- ❖ Low dose may be used as sleep aid with no anticholinergic effects

NaSSAs

1. Mianserin (Tolvon[®])
2. Mirtazapine (Remeron[®] SOLTAB)





8. Miscellaneous

Class	Drugs	Mechanism of action	Comments
Melatonergic agonist and serotonergic antagonist	Agomelatine (Valdoxan [®])	▪ Agonist of MT₁ and MT₂ ▪ Block 5-HT_{2A}, 5-HT_{2C}	Low rate of adverse reactions including weight gain
Norepinephrine reuptake inhibitors (NRIs)	Reboxetine (Edronax [®])	▪ Block NET	Sympathetic and activating side effects are of concern
Serotonin partial agonist reuptake inhibitors (SPARIs)	Vilazodone (Viibryd [®])	▪ Block 5-HTT ▪ 5-HT_{1A} partial agonist	
Multimodal antidepressant	Vortioxetine (Brintellix [®])	▪ Block 5-HTT ▪ 5-HT_{1A} partial agonist ▪ 5-HT₃, 5-HT₆ and 5-HT₇ antagonist	Multiple mechanisms contributing to improved cognitive domains in MDD
NMDA receptor antagonist	Ketamine (low dose)	▪ Block NMDA receptor and change glutamate neurotransmission ▪ Activate mTOR ▪ Induce rapid synaptic formation	Rapid antidepressant action within 24 hours after iv infusion in patient with treatment resistant MDD

Other adverse effects

- ✧ Weight gain
 - ❖ Most likely- TCA, mirtazapine>>>trazodone
- ✧ QTc interval prolongation
 - ❖ Most likely- TCAs
- ✧ Lower seizure threshold
 - ❖ Most likely- TCAs, maprotiline, bupropion
 - ❖ Less likely- MAOIs, SSRIs
- ✧ Hyponatremia- esp. elderly
 - ❖ SSRI/SNRI
- ✧ Increase risk of bleeding
 - ❖ SSRI/SNRI

Antidepressants	Major elimination pathway	Other elimination pathways
TCA	CYP2D6, CYP2C19	CYP3A4, CYP1A2
SSRI		
Citalopram	CYP2C19	CYP3A4, CYP2D6
Escitalopram	CYP2C19	CYP3A4, CYP2D6
Fluoxetine (active metabolite norfluoxetine)	CYP2D6	CYP2C9, CYP2C19, CYP3A4
Fluvoxamine	CYP1A2, CYP2D6	
Paroxetine	CYP2D6	CYP3A4
Sertraline	CYP2B6	CYP2C19, CYP2C9, CYP3A4, CYP2D6
SNRI		
Desvenlafaxine	Renal	UGT, CYP3A4
Duloxetine	CYP2D6, CYP1A2	
Milnacipran	Renal	UGT, CYP3A4
Venlafaxine	CYP2D6	CYP3A4
Others		
Agomelatine	CYP1A2	CYP2C9, CYP2C19
Bupropion (active metabolite hydroxybupropion)	CYP2B6	
Mirtazapine	CYP2D6, CYP3A4	CYP1A2, UGT
Reboxetine	CYP3A4	
Vilazodone	CYP3A4	CYP2C19, CYP2D6, carboxylesterase
Vortioxetine	CYP2D6	CYP3A4, CYP2C19, CYP2C9, CYP2A6, CYP2C8, CYP2B6
Trazadone	CYP3A4	CYP2D6

Common drug interactions with antidepressants

✧ Pharmacodynamic drug interactions

❖ Multiple serotonergic drugs → serotonin syndrome

- ✧✧ SSRIs, SNRIs
- ✧✧ Tramadol, pethidine
- ✧✧ Ergotamine, triptan
- ✧✧ MAO-inhibitor - linezolid

❖ TCAs + cholinergic drugs

- ✧✧ TCA + bethanechol
- ✧✧ TCA + cholinesterase inhibitors

✧ Pharmacokinetic drug interactions

❖ Many antidepressants inhibit CYP450 isozymes

- ✧✧ Significant CYP2D6 inhibitors: fluoxetine, paroxetine, bupropion, duloxetine
- ✧✧ Fluoxetine inhibits most CYPs except CYP1A2
- ✧✧ Fluvoxamine inhibits most CYPs except CYP2D6

บัญชียาหลักแห่งชาติ พ.ศ. 2569

4.3 Antidepressant drugs

1.	Amitriptyline hydrochloride	TCA	tab	b
2.	Fluoxetine hydrochloride	SSRI	cap, tab	b
3.	Imipramine hydrochloride	TCA	tab	b
4.	Mianserin hydrochloride	NaSSA	tab	b
5.	Nortriptyline hydrochloride	TCA	tab	b
6.	Sertraline	SSRI	tab (เฉพาะ 50 mg)	b
7.	Clomipramine hydrochloride	TCA	cap, tab	s
8.	Trazodone hydrochloride	SARI	tab	s
9.	Escitalopram oxalate	SSRI	tab (เฉพาะ 10, 20 mg)	ex

เงื่อนไข

ใช้เป็น antidepressant ที่เป็ยยาหลัก ในการรักษา treatment-resistant depression (TRD) โดยมีเกณฑ์ดังนี้

1. ได้รับการวินิจฉัย และการรักษาเบื้องต้นโดยแพทย์ผู้เชี่ยวชาญด้านจิตเวชศาสตร์
2. ใช้ antidepressant อื่นมาแล้วอย่างน้อย 2 ชนิด ในระยะเวลาและขนาดยาที่เหมาะสมแล้ว ยังควบคุมอาการไม่ได้

Clinical applications of antidepressants

- ✧ Major depressive disorder
- ✧ Anxiety disorders
 - ❖ SSRIs, venlafaxine, duloxetine
- ✧ PTSD
 - ❖ SSRI
- ✧ OCD
 - ❖ Clomipramine and SSRI
- ✧ Eating disorders
 - ❖ Binge eating disorder; SSRIs
 - ❖ Bulimia nervosa; SSRIs, TCAs
- ✧ Insomnia as sleep aid
 - ❖ Trazodone, mirtazapine, agomelatine, TCA
- ✧ Smoking cessation aid
 - ❖ Bupropion, nortriptyline
- ✧ Premenstrual dysphoric disorder
 - ❖ SSRIs
- ✧ Neuropathic pain and fibromyalgia
 - ❖ SNRIs, TCAs
- ✧ Vasomotor symptoms-associated with menopause
 - ❖ SSRIs, desvenlafaxine
- ✧ Premature ejaculation
 - ❖ SSRIs
- ✧ Migraine prevention
 - ❖ Amitriptyline

Lithium

- ✿ An alkaline metal monovalent cation similar to sodium
- ✿ A mood stabilizer for bipolar disorder
 - ✦ Drugs that reduce severity and shorten the duration of mood episodes in bipolar disorder



LITHIUM
CARBONATE
300 MG.



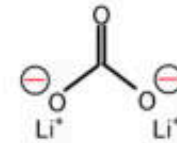
LITHIUM
CARBONATE
300 MG.

ACS
Chemistry for Life®

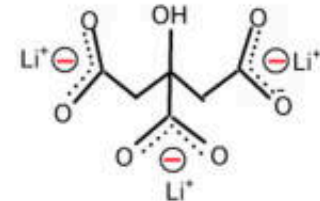
PERIODIC TABLE OF ELEMENTS

American Chemical Society

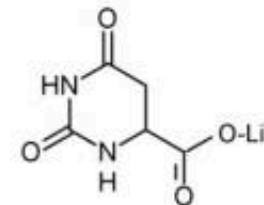
www.acs.org/outreach



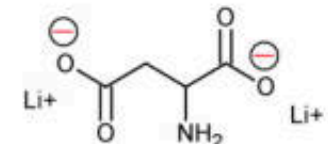
Lithium carbonate



Lithium citrate



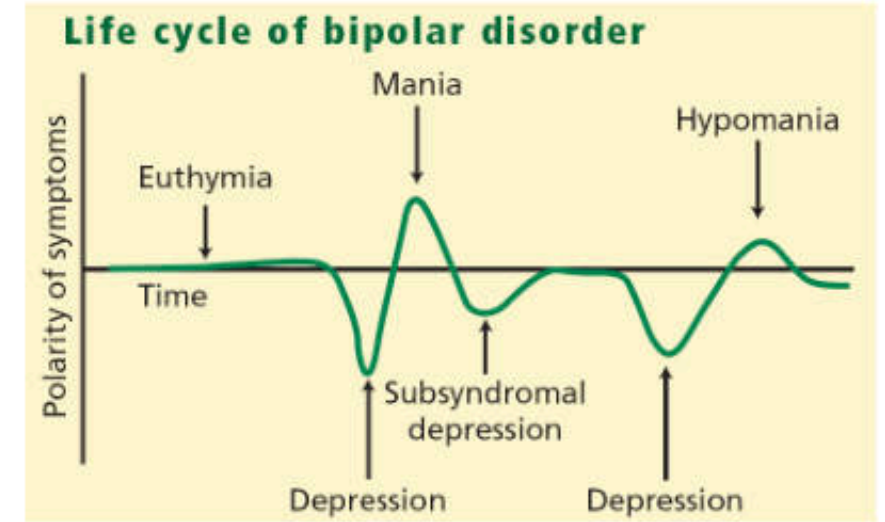
Lithium orotate



Lithium aspartate

Lithium

- ✧ Major clinical uses
 - ❖ Treatment of manic episode in bipolar disorder
 - ❖ Prevention of recurrent manic and depressive episodes
 - ❖ Rapid cycling
 - ❖ Prevention of suicide in bipolar disorder
- ✧ Narrow therapeutic index
 - ❖ Therapeutic level 0.6-1.2 mEq/L



High Alert Drug

Lithium mechanisms of action

- ✧ Widely alter neuronal function
 - ✦ Neurotransmitter levels/functions
 - ✦ Intracellular signalling
- ✧ Has both antimanic and anti-depressive effect

System	Effect of lithium
5-HT (serotonin) function	Greatly increased
Acetylcholinesterase function	Greatly increased
Sodium function	Increased
Dopamine function	Reduced
GABA function	Increased
Inositol	Reduced
cAMP	Reduced
Protein kinase C	Reduced
Glycogen synthase kinase-3 β (GSK-3 β)	Greatly reduced
BDNF	Increased
Bcl-2	Increased
Pro-apoptotic proteins (p53, BAX)	Reduced

Pharmacokinetics

- ✧ Well absorbed to systemic and brain tissue
- ✧ Monovalent cation without protein binding
- ✧ Almost excreted unchanged in urine (95%)
 - ❖ Depends on renal blood flow

ADRs of lithium

✧ GI

- ❖ Nausea/ vomiting (2-3 first week)

✧ CNS

- ❖ Fine tremor (15-53%)
- ❖ Slow thinking, memory problem
- ❖ Somnolence
- ❖ Weight gain 20% gain more than 10 kg

✧ Renal

- ❖ Unable to concentrate urine → polyuria, polydipsia
- ❖ Diabetes insipidus

✧ Endocrine

- ❖ Hypothyroidism

✧ CVS

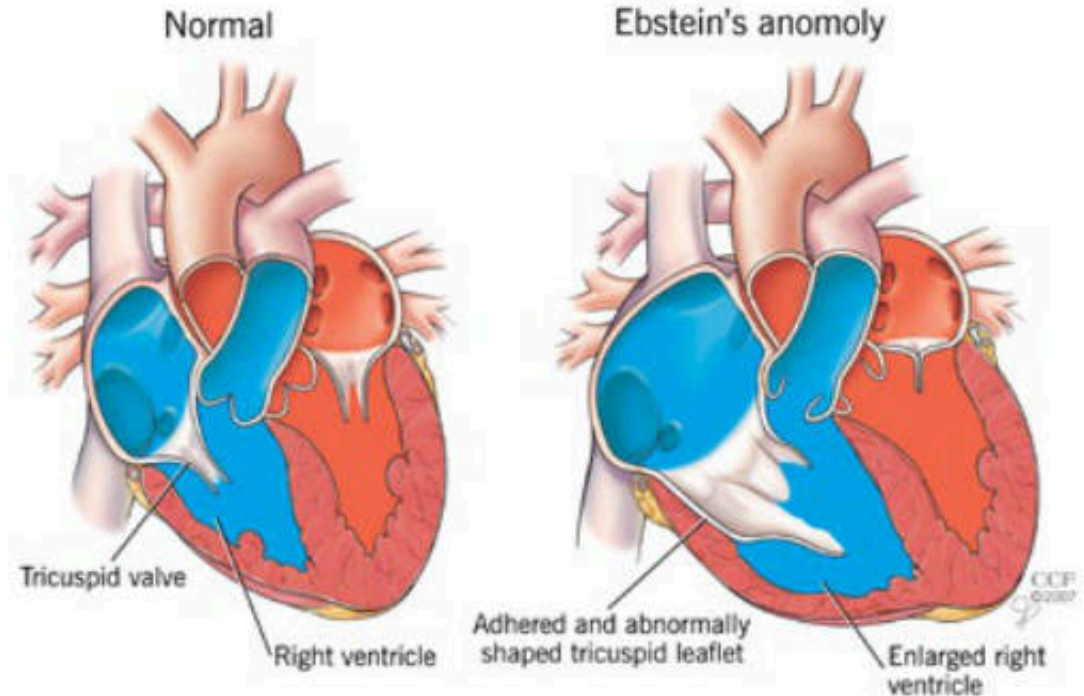
- ❖ Cardiac T-wave inversion

✧ Cutaneous

- ❖ Pruritic, maculopapular rash
- ❖ May appear during first month of treatment

Lithium in pregnancy and lactation

- ✧ Should be avoided in pregnant and breast-feeding women
- ✧ Cardiac malformation (Ebstein's anomaly) at 0.05 – 0.1% in the first trimester
- ✧ Risk of infant toxicity



Pharmacokinetic drug interactions of lithium

✧ Drugs that change glomerular blood flow

	Glomerular blood flow	Blood lithium level
NSAIDs	↓	↑
ACEIs, ARBs	↓	↑
Theophylline, caffeine	↑	↓

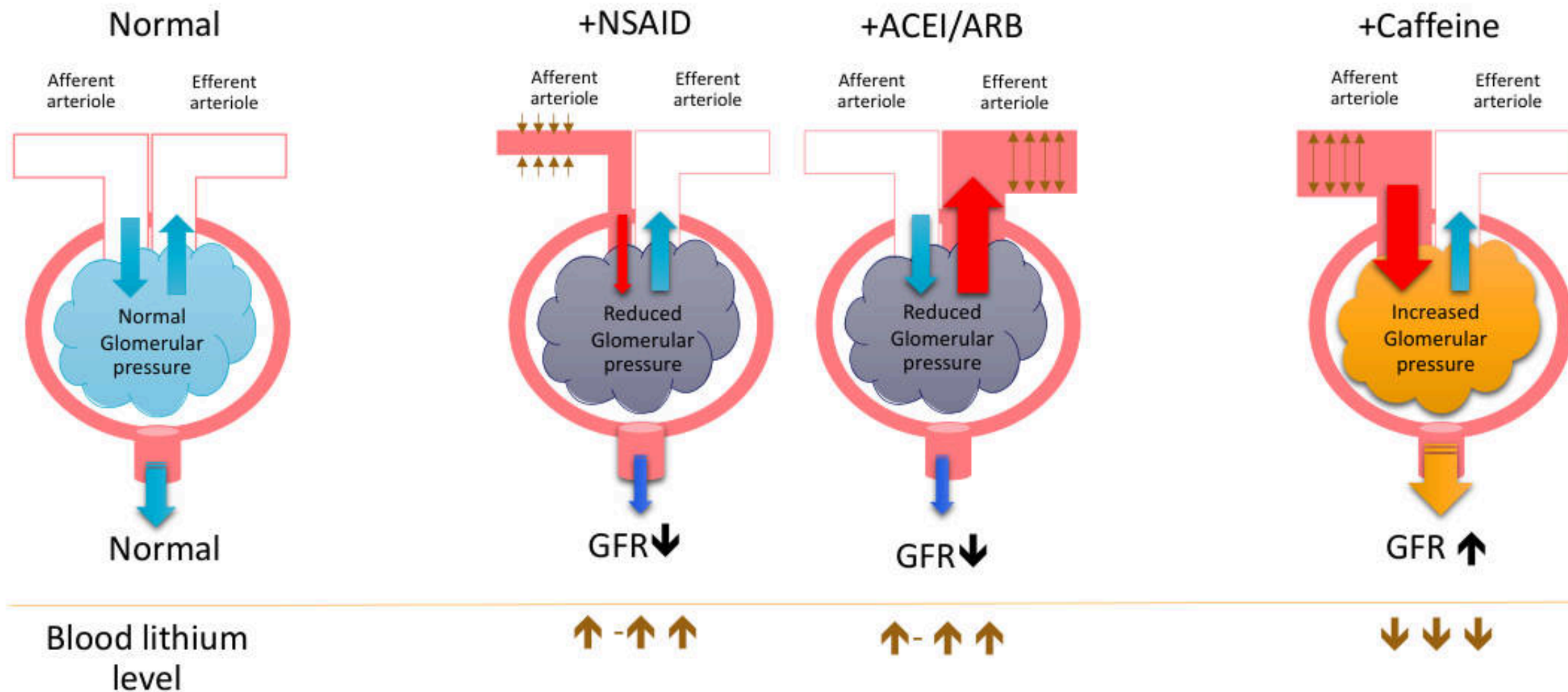
✧ Drugs that change lithium reabsorption

❖ Diuretics; hydrochlorothiazide, loop diuretics

✧✧ Increase Na^+ , Li^+ excretion → Increase Na^+ , Li^+ reabsorption

❖ Dehydration

✧✧ Increase Na^+ , Li^+ reabsorption



Impact on lithium concentrations: ↑ = increased 0–20 %; ↑ ↑ = increased 20–40 %; ↓ ↓ ↓ decrease 40–60 %)